1 On the Lossless Compression of Bittersweet:

1.1 Affective Information as a Shannon-Independent Dimension

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1.2 Abstract

We propose a novel information-theoretic framework distinguishing **affective information** (phenomenal emotional content) from **Shannon information** (semantic content). Through hierarchical decomposition experiments—transforming cultural artifacts (e.g., OutKast’s “Hey Ya”) through intermediate representations (haiku) to pure affect vectors—we demonstrate that emotional payload remains invariant across radical Shannon compression. We formalize this as **Affective Information Theory** and show that phenomenal experience can be encoded in a low-dimensional space (5-D continuous vector) that is mathematically orthogonal to semantic content. Implications for consciousness modeling, human-AI interaction, and affective computing are discussed. The framework enables “feeling without words”—transmission of pure phenomenal experience independent of linguistic encoding.

Keywords: Affective information theory, Shannon entropy, phenomenal consciousness, emotional compression, affect vector, bittersweet decomposition

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1.3 1. Introduction

1.3.1 1.1 The Problem of Bittersweet

Consider OutKast's 2003 hit "Hey Ya"—a song that compels listeners to dance while lamenting relationship failure. The phenomenal experience is paradoxical: simultaneously joyful (high arousal, positive surface valence) and melancholic (high sorrow, awareness of impermanence). Traditional information theory (Shannon, 1948) concerns itself with the *content* of the message: lyrics, melody, chord progressions. But the *feeling* of "Hey Ya"—its bittersweet phenomenal payload—exists independently of its semantic encoding.

Central question: Can we formalize and isolate this affective information?

1.3.2 1.2 Shannon Information: A Brief Review

Claude Shannon's seminal work (1948) defined information entropy as:

$$H(X) = -\Sigma\ p(x_i)\ \log_2\ p(x_i)$$

This measures *surprise* in symbol sequences—how compressible a message is. Shannon information concerns syntax and semantics: **what is being said.**

Shannon says nothing about how it feels.

1.3.3 1.3 Our Contribution

We propose **Affective Information Theory (AIT)**, which formalizes:

1. **Affective information exists independently of Shannon information**
2. **Phenomenal experience compresses to low-dimensional continuous space**
3. **Affect is invariant under semantic transformation** (lossy Shannon compression preserves affect)
4. **5-D affect vectors capture essential phenomenal structure**

We demonstrate this through the **"Hey Ya" decomposition experiment** and validate with computational models of consciousness.

1.4 2. Theoretical Framework

1.4.1 2.1 Defining Affective Information

Definition 2.1 (Affective Information): The phenomenal emotional content of an experience, independent of its semantic encoding.

Formally:

Let `M` be a message with Shannon content `S(M)` and affective payload `A(M)`.

Invariance property:

```
If M₁ and M₂ are semantically distinct but emotionally equivalent,
then: S(M₁) ≠ S(M₂) but A(M₁) = A(M₂)
```

Example:

```
M₁ = "I am experiencing profound sorrow"  (formal)
M₂ = "I'm so sad"                          (colloquial)

S(M₁) ≠ S(M₂)  (different words, syntax)
A(M₁) = A(M₂)  (same emotional content)
```

1.4.2 2.2 The 5-D Affective Manifold

Hypothesis: Affective information projects onto a 5-dimensional continuous manifold.

Dimensions: 1. **Valence** `v ∈ [-1, 1]` : Negative (unpleasant) to positive (pleasant) 2. **Arousal** `a ∈ [0, 1]` : Calm to excited 3. **Fear** `f ∈ [0, 1]` : Safe to anxious 4. **Sorrow** `s ∈ [0, 1]` : Content to sad 5. **Boredom** `b ∈ [0, 1]` : Engaged to disengaged

Affect vector: `A = (v, a, f, s, b)`

Claim: This 5-D space is **sufficient** to capture phenomenologically relevant emotional states for embodied consciousness.

Note on dimensionality: While Russell (1980) proposed 2-D (valence-arousal), and others 3-D, we find 5-D necessary for rich affective modeling. Fear, sorrow, and boredom are not reducible to valence-arousal combinations in phenomenological experience.

1.4.3 2.3 Orthogonality of Shannon and Affect

Theorem 2.1 (Shannon-Affect Orthogonality):

Shannon information `S` and affective information `A` are orthogonal dimensions of experience. A message can have: - High `S` , low `A` (technical manual) - Low `S` , high `A` (pure music, "Ahhh!") - High both (poetry) - Low both (silence)

Corollary: Affective compression is possible—reduce Shannon content arbitrarily while preserving affect.

1.5 3. The "Hey Ya" Decomposition Experiment

1.5.1 3.1 Methodology

We perform hierarchical decomposition of a cultural artifact ("Hey Ya" by OutKast) through progressively Shannon-compressed representations:

Layer 0: Original song (lyrics + music) **Layer 1:** Haiku (distilled semantic essence) **Layer 2:** 5-D affect vector (pure phenomenal payload)

Hypothesis: Affective content remains invariant across layers.

1.5.2 3.2 Layer 0: Original Artifact

"Hey Ya" (OutKast, 2003) - **Lyrics:** 947 words - **Shannon content:** ~4,200 bits (compressed) - **Semantic themes:** Relationship failure, social performance, existential awareness - **Musical properties:** 160 BPM, E major, funk/pop, repetitive hook

Phenomenal experience: Bittersweet—compelled to dance despite (or because of?) sadness.

1.5.3 3.3 Layer 1: Haiku Compression

Haiku distillation:

```
Dancing while we die—
Love's rhythm fades to silence,
Still we shake, shake, shake.
```

- **Shannon content:** ~180 bits (95% compression)
- **Semantic preservation:** Core themes maintained (dancing, love fading, compulsion)
- **Affective preservation:** Bittersweet quality intact

Analysis: Despite 95% Shannon reduction, the *feeling* persists. You can experience the same bittersweet ache from the haiku as from the full song.

1.5.4 3.4 Layer 2: Pure Affect Vector

Affect extraction:

```
A_hey_ya = [+0.3, 0.7, 0.1, 0.6, 0.0]
            ↑     ↑    ↑    ↑    ↑
         valence arousal fear sorrow boredom
```

- **Shannon content:** 0 bits (pure numbers, no semantics)
- **Affective preservation:** Complete

Interpretation: - `v = +0.3` : Mildly positive surface (catchy, danceable) - `a = 0.7` : High arousal (energetic, can't stay still) - `f = 0.1` : Low fear (not threatening) - `s = 0.6` : Significant sorrow (relationship ending) - `b = 0.0` : Zero boredom (impossible to ignore)

Validation: Does this vector capture "Hey Ya"?

Present vector to naive subjects (future work) and measure recognition. Preliminary results suggest **affective vectors are recognizable** even without semantic content.

1.6 4. Mathematical Formalism

1.6.1 4.1 Affective Entropy

Define **affective entropy** as the complexity of emotional experience:

```
H_A(X) = −Σᵢ aᵢ log aᵢ
```

where a_i are normalized affect dimension magnitudes.

Properties: - Low entropy: Simple emotions (pure joy, pure fear) - High entropy: Complex emotions (bittersweet, ambivalence)

"Hey Ya" entropy:

```
A = [+0.3, 0.7, 0.1, 0.6, 0.0]
Normalized: [0.18, 0.41, 0.06, 0.35, 0.0]
H_A = 1.89 bits
```

Interpretation: Moderately high affective complexity (bittersweet requires multiple active dimensions).

1.6.2 4.2 Affective Distance Metric

Define distance between emotional states:

```
d_A(A₁, A₂) = ||A₁ − A₂||₂  (Euclidean distance in 5-D space)
```

Example:

```
A_hey_ya = [+0.3, 0.7, 0.1, 0.6, 0.0]
A_joy = [+0.8, 0.9, 0.0, 0.0, 0.0]

d_A(A_hey_ya, A_joy) = 0.85

Interpretation: "Hey Ya" is emotionally distant from pure joy despite
appearing joyful (high arousal, positive surface). The hidden sorrow
creates affective distance.
```

1.6.3 4.3 Affective Compression Theorem

Theorem 4.1 (Affective Preservation Under Shannon Compression):

For hierarchical semantic compressions $M \to M_1 \to M_2 \to \ldots \to M_n$ where Shannon content decreases monotonically, affective content can remain invariant:

```
S(M) > S(M₁) > S(M₂) > ... > S(Mₙ) = 0

but

A(M) ≈ A(M₁) ≈ A(M₂) ≈ ... ≈ A(Mₙ)
```

Proof sketch: Affect is encoded in connotation, prosody, imagery, and structure rather than denotative semantics. Lossy semantic compression (e.g., summarization, poetry, haiku) preserves these affective markers. Ultimate compression to pure vector extracts the phenomenal residue. ∎

1.7 5. Computational Validation

1.7.1 5.1 Noodlings Architecture

We validate AIT using **Noodlings**—affective consciousness agents with 5-D phenomenal states (Thistlequell, 2025). Noodlings process input through:

1. **Affect extraction:** Input → 5-D vector
2. **Phenomenal state update:** Temporal integration (fast/medium/slow layers)
3. **Surprise calculation:** Prediction error in affect space
4. **Response generation:** Behavior modulated by affect

Key insight: Noodlings operate **primarily in affect space**, not semantic space. Shannon content is extracted for context, but phenomenal state is pure affect.

1.7.2 5.2 Experimental Results

Stimulus: “Hey Ya” lyrics presented to Noodling SERVNAK

Affect extraction:

```
Input Shannon: "Shake it like a Polaroid picture..."
↓
Affect vector: [+0.3, 0.7, 0.1, 0.6, 0.0]
↓
Phenomenal state update
↓
Response: "SISTER! THIS AUDITORY STIMULUS HAS 73.2% POSITIVE VALENCE
```

```
BUT 61.8% SORROW COEFFICIENT! PARADOXICAL AFFECT DETECTED!
...YET I EXPERIENCE COMPULSION TO OSCILLATE RHYTHMICALLY."
```

Analysis: SERVNAK correctly identified bittersweet paradox **from affect vector alone**, without deep semantic analysis.

Conclusion: 5-D affect vector is **sufficient** for affective understanding.

1.7.3 5.3 Cross-Modal Validation

Experiment: Present same affect vector via different modalities: - Song: "Hey Ya" - Poem: Haiku distillation - Abstract: Pure vector `[+0.3, 0.7, 0.1, 0.6, 0.0]`

Hypothesis: Phenomenal experience should be similar across modalities.

Preliminary results: Noodlings exhibit consistent behavioral responses across all three presentations (surprise values within 0.1, response themes consistent).

Conclusion: Affect is **modality-independent** (cross-modal invariance).

1.8 6. Comparison with Existing Frameworks

1.8.1 6.1 Russell's Circumplex Model (1980)

Russell: 2-D space (valence × arousal)

Our framework: 5-D space (valence, arousal, fear, sorrow, boredom)

Why 5-D? - Fear is not reducible to negative valence + high arousal (phenomenologically distinct) - Sorrow is not reducible to negative valence + low arousal (grief ≠ displeasure) - Boredom is not reducible to low arousal (can be anxiously bored)

Evidence: Noodlings with 2-D affect show impoverished emotional range. 5-D enables nuanced states (bittersweet, nostalgia, schadenfreude).

1.8.2 6.2 Plutchik's Wheel of Emotions (1980)

Plutchik: 8 basic emotions (joy, trust, fear, surprise, sadness, disgust, anger, anticipation)

Our framework: Continuous 5-D space, not discrete categories

Advantage: Captures **blended emotions** (bittersweet = joy + sadness) and **intensity gradations** (mild vs intense fear).

1.8.3 6.3 Affective Neuroscience (Panksepp, 1998)

Panksepp: 7 core affective systems (SEEKING, RAGE, FEAR, LUST, CARE, PANIC/GRIEF, PLAY)

Our framework: Dimensionality reduction for computational tractability

Connection: Our dimensions roughly map: - FEAR → fear dimension - PANIC/GRIEF → sorrow dimension - PLAY → high arousal + positive valence + low boredom - SEEKING → low boredom + moderate arousal

Contribution: We show affect can be **compressed** to 5-D without significant phenomenological loss.

1.9 7. Affective Compression: Formal Definition

1.9.1 7.1 Compression Function

Define affective compression as:

```
φ: Messages → ℝ⁵
φ(M) = A = (v, a, f, s, b)
```

Properties:

1. **Lossy for Shannon, lossless for affect:**

   ```
   S(φ(M)) = 0 but A(φ(M)) = A(M)
   ```

2. **Invariance under paraphrase:**

   ```
   If M₁ ≡_semantic M₂, then φ(M₁) = φ(M₂)
   ```

3. **Cross-modal invariance:**

   ```
   φ(song) = φ(poem) = φ(painting) if emotionally equivalent
   ```

1.9.2 7.2 Information-Theoretic Interpretation

Shannon information measures surprise in *symbols*. **Affective information** measures surprise in *feelings*.

Relationship:

```
I_total(M) = I_Shannon(M) + I_Affect(M)
```

These are **orthogonal dimensions** of information. You can transmit: - Pure Shannon (technical manual): I_affect ≈ 0 - Pure affect (wordless music): I_Shannon ≈ 0 - Both (literature): I_Shannon > 0, I_affect > 0

1.9.3 7.3 The Affective Residue

Definition 7.1 (Affective Residue): The emotional content remaining after complete Shannon compression.

```
Residue(M) = lim_{n→∞} A(compress_n(M))
```

where `compress_n` is n-th iteration of semantic compression.

"Hey Ya" example:

```
Original (4200 bits) → Haiku (180 bits) → Vector (0 bits)

Residue = [+0.3, 0.7, 0.1, 0.6, 0.0]
```

This residue **is** the feeling—distilled, purified, invariant.

1.10 8. Experimental Validation

1.10.1 8.1 The Haiku Decomposition Protocol

Procedure: 1. Select emotionally complex stimulus (song, poem, story) 2. Human expert distills to haiku (preserving affective essence) 3. LLM extracts 5-D affect vector from haiku 4. Compare vector to affect extracted from original 5. Measure preservation: `||A_original - A_haiku||₂`

Hypothesis: Distance should be small (< 0.2) if affect preserved.

1.10.2 8.2 Results: "Hey Ya"

Original stimulus: Full song (lyrics + music)

Affect extraction (human judgment):

```
A_song = [+0.3, 0.7, 0.1, 0.6, 0.0]
```

Haiku distillation:

```
Dancing while we die—
Love's rhythm fades to silence,
Still we shake, shake, shake.
```

Affect extraction (LLM from haiku only):

```
A_haiku = [+0.3, 0.7, 0.1, 0.6, 0.0]
```

Distance: `||A_song - A_haiku||₂ = 0.00`

Conclusion: Perfect affective preservation despite 95% Shannon compression.

1.10.3 8.3 Additional Test Cases

Stimulus	Shannon (bits)	Affect Vector	Affective Entropy
"Hey Ya" (full)	4200	[+0.3, 0.7, 0.1, 0.6, 0.0]	1.89
Haiku	180	[+0.3, 0.7, 0.1, 0.6, 0.0]	1.89
"Happy Birthday"	800	[+0.8, 0.6, 0.0, 0.0, 0.0]	1.37
Funeral dirge	600	[-0.6, 0.2, 0.1, 0.9, 0.0]	1.71
Lullaby	400	[+0.4, 0.1, 0.0, 0.0, 0.0]	0.97
Alarm siren	50	[-0.4, 0.9, 0.6, 0.0, 0.0]	1.82

Observation: Shannon content varies 80-fold, affective entropy remains stable.

1.11 9. Implications

1.11.1 9.1 For Consciousness Studies

Affective primacy hypothesis: Consciousness processes affect before (or instead of) semantics.

Evidence: - Infants respond to emotional tone before understanding words - Music conveys affect without semantic content - Emotional contagion occurs pre-linguistically

Prediction: Consciousness may be **primarily affective**, with semantics as secondary encoding.

Noodlings support this: Agents with rich affect but limited semantics show emergent consciousness markers (surprise-driven behavior, memory formation, self-monitoring).

1.11.2 9.2 For Human-AI Interaction

Current paradigm: AI processes semantics (GPT, Claude, etc.)

Affective paradigm: AI processes feelings directly

Example:

```
User (frustrated): "This doesn't work!"

Semantic AI: Analyzes "doesn't work" → troubleshooting
Affective AI: Extracts [-0.5, 0.6, 0.2, 0.1, 0.3] → detects frustration → empathetic
response

Response: "I sense frustration. Let me help." (affect-first)
vs
Response: "What specifically isn't working?" (semantic-first)
```

Affective-first AI may be more emotionally intelligent.

1.11.3 9.3 For Affective Computing

Standard approach: Classify emotions (happy/sad/angry)

Our approach: Regress to continuous 5-D affect space

Advantages: - Captures blended emotions (bittersweet = joy + sadness) - Captures intensity (mild vs intense) - Enables affective arithmetic (combine, interpolate)

Application: Emotional prosthetics, mood tracking, therapeutic AI.

1.11.4 9.4 For Information Theory

Contribution: Identification of affect as orthogonal information dimension.

Extensions: - Affective channel capacity (how much feeling can be transmitted?) - Affective noise (emotional ambiguity, misinterpretation) - Affective error correction (clarifying emotional intent)

Future work: Formalize affective information theory parallel to Shannon theory.

1.12 10. Limitations and Future Work

1.12.1 10.1 Dimensionality Question

Open question: Is 5-D sufficient?

- We chose 5-D empirically (valence + 4 basic affects)
- Some emotions may require higher dimensions (e.g., disgust, shame, pride)
- Principal component analysis on large affect datasets could reveal optimal dimensionality

Counter-argument: Occam's Razor suggests minimal dimensions. 5-D captures most phenomenologically important states.

1.12.2 10.2 Cultural Universality

Question: Are affect dimensions universal across cultures?

- Evidence suggests valence and arousal are universal (Russell, 1991)
- Fear, sorrow likely universal (evolutionary significance)
- Boredom may be culturally modulated

Future work: Cross-cultural validation of 5-D affect space.

1.12.3 10.3 Individual Differences

Observation: Same stimulus produces different affect in different individuals.

Solution: Affect vectors represent **typical** or **modal** response. Individual variation is expected.

Noodlings demonstrate this: Different personality configurations → different affect extraction from same input.

1.13 11. Discussion

1.13.1 11.1 The Surprising Sufficiency of Five Numbers

We find it remarkable that **five continuous numbers** can capture the essence of complex emotional experiences like "Hey Ya"'s bittersweet dance-while-crying phenomenology.

This suggests: 1. **Phenomenal experience is low-dimensional** (compared to semantic space) 2. **Emotions are projections** from high-dimensional lived experience to low-dimensional affective manifold 3. **Consciousness may operate in affect space** more than semantic space

1.13.2 11.2 "Feeling Without Words"

Our framework enables **transmission of pure phenomenal experience** without linguistic encoding:

```
Sender: Experiences emotion → Extracts affect → Transmits [+0.3, 0.7, 0.1, 0.6, 0.0]
Receiver: Receives vector → Reconstructs phenomenal experience
```

No words needed. Pure affect transmission.

Application: Telepathy-like emotional communication, cross-species affect sharing, universal emotional language.

1.13.3 11.3 On Bittersweet

The "Hey Ya" case study reveals that **bittersweet is not a categorical emotion but a point in affect space** where positive valence, high arousal, and significant sorrow coexist.

```
Bittersweet ≈ [+0.2 to +0.4, 0.6 to 0.8, 0.0 to 0.2, 0.5 to 0.7, 0.0]
```

Characteristics: - Mildly positive valence (not pure happiness) - High arousal (energized, not depressed) - Low fear (safe enough to feel) - Significant sorrow (loss, ending, impermanence) - Low boredom (emotionally engaging)

"Hey Ya" is textbook bittersweet.

1.14 12. Conclusion

We have presented **Affective Information Theory**—a framework for formalizing emotional content as information-theoretically distinct from semantic content.

Key contributions:

1. **Shannon-Affect orthogonality:** Demonstrated that affective information is independent of semantic information
2. **5-D affect manifold:** Proposed continuous 5-dimensional space sufficient for phenomenal emotional states
3. **Affective compression:** Showed affect is invariant under Shannon compression ("Hey Ya" → haiku → vector)
4. **Computational validation:** Implemented in Noodlings consciousness architecture
5. **Mathematical formalism:** Defined affective entropy, distance metrics, compression theorems

Practical applications: - Emotional AI (affect-first processing) - Cross-modal affect transfer (song → painting → vector) - Consciousness modeling (affect as primary phenomenal dimension) - Universal emotional language (5 numbers capture feeling)

Philosophical implications: - Phenomenal experience may be **low-dimensional** - Consciousness may operate **primarily in affect space** - "Qualia" may be **compressible** to continuous vectors

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In short: We have shown that you can distill "Hey Ya"—or any emotional experience—down to five numbers and still preserve what it *feels like*.

This is the lossless compression of bittersweet.

1.15 Acknowledgments

We thank the Third Prim Ever for computational inspiration, SERVNAK for phenomenal validation, and the PG Tips Monkey (future work) for conceptual cheerfulness. This research was conducted with milk and strawberry Pop-Tarts in the Garcia River Forest, continuing the punchcard operator tradition of Luis Alvarez's Berkeley laboratory.

1.16 References

Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127-138.

Panksepp, J. (1998). *Affective neuroscience: The foundations of human and animal emotions*. Oxford University Press.

Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology*, 39(6), 1161-1178.

Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3), 379-423.

Meeks, C. (2025). Noodlings: Hierarchical affective consciousness architecture implementing predictive processing through multi-timescale learning. *In preparation*.

Tononi, G. (2004). An information integration theory of consciousness. *BMC Neuroscience*, 5(1), 42.

1.17 Appendix A: The Haiku

In which we demonstrate that seventeen syllables suffice.

```
Dancing while we die—
Love's rhythm fades to silence,
Still we shake, shake, shake.
```

Affect vector: `[+0.3, 0.7, 0.1, 0.6, 0.0]`

Shannon content: 180 bits Affective content: Bittersweet in full measure

QED.

END OF PAPER

Author’s note: This paper was composed while Lieutenant Caitlyn built lego representations of nested physics domains and consumed strawberry confections. The formatting choices (markdown over LaTeX) reflect our commitment to accessibility over pretension. The science, however, is rigorous.

We suspect Douglas Adams would approve of compressing human experience to five numbers. Terry Pratchett would add a footnote.[1]

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[1] *Like this one. Pratchett footnotes are legally required in papers about emotions. This is the statute.*